15.095: Machine Learning Under a Modern Optimization Lens **Fall 2026**

Homework 3: Due October 19

Hand in: **pdf** upload to Gradescope. Please append any Julia code **at the end** of the whole pdf.

Note: this homework covers lectures 8–11. We recommend attempting questions after the relevant content has been covered in class. Total: 100 points.

Question 1: True/False (30 points)

Please classify the following statements as true or false and justify your answer. We will assign 1 point for correctly classifying the answer, and 5 points for the validity of the justification/counterexample.

(a) (6 points) Suppose we train a random forest in which every tree is allowed to consider all p features at every split, rather than a random subset of them. Then all the trees in the forest are identical, and the forest makes the same predictions as a single CART tree grown the same way.

(b) (6 points) Suppose feature j takes strictly positive values, and we replace x_{ij} by $\log x_{ij}$ for every $i \in [n]$, leaving the other features unchanged. For a fixed maximum depth, minimum leaf size $N_{\min}$ and complexity parameter α, an OCT (with parallel splits) trained to optimality on the transformed data attains the same optimal objective value as an OCT trained to optimality on the original data.

(c) (6 points) In the OCT mixed-integer formulation, recall that the binary variable c_{kt} equals 1 if leaf t predicts class k, and z_{it} equals 1 if point i is assigned to leaf t. Suppose that, on top of the minbucket constraint, we require every leaf that predicts class A to contain at least N_A training points. This requirement can be added to the formulation using one linear constraint per leaf and no additional variables.

(d) (6 points) A bakery decides each morning how many loaves $z \geq 0$ to bake. Each loaf costs \$7 to bake and sells for \$10, and loaves not sold by closing are thrown away, so if demand turns out to be y the day's cost is

$$c(z; y) = 7z - 10 \min(z, y).$$

The bakery has trained a CART tree on past days, with features such as the day of the week and the weather forecast, to predict demand. Tomorrow falls in a leaf containing five past days, with demands of 40, 50, 60, 70 and 130 loaves. Then the weighted-cost prescription from lecture, using the CART weights, bakes as many loaves as the tree predicts, namely 70.

(e) (6 points) We train an Optimal Policy Tree on observational data with $m \geq 2$ treatments, where point i received treatment T_i and has outcome $y_i \in \mathbb{R}$, which we want to maximize. The rewards are estimated with inverse propensity weighting,

$$\Gamma_{it} = \mathbb{1}\{T_i = t\} \cdot \frac{y_i}{p_{it}}, \qquad i \in [n],\ t \in [m],$$

where $p_{it} > 0$ is the estimated propensity score. Suppose the tree is deep enough to place every training point in its own leaf. Then the tree prescribes to every training point the treatment that point actually received.

Question 2: Greedy and Optimal Trees (40 points)

An emergency department wants a short rule for deciding whether a patient who arrives with chest pain should be admitted (class A) or sent home with a follow-up appointment (class B). Two measurements are available. The first, x_1, is a bedside score the triage nurse can compute in a few minutes. The second, x_2, comes from a lab panel that takes an hour to return and has to be paid for. Both have been rescaled to lie in $(0, 3)$.

The department has records of $n = 31$ past patients with known outcomes. Figure 3.1 reports how many patients of each class fall in each unit cell of the (x_1, x_2) plane; no patient lies on a grid line.

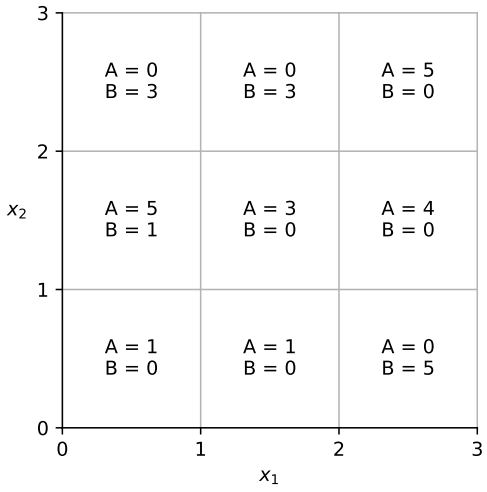


Figure 3.1: Number of patients of class A and class B in each cell.

Throughout, a tree has depth at most 2, and every split has the form $x_j < b$ with $j \in \{1, 2\}$ and an integer threshold $b \in \{1, 2\}$; patients with $x_j < b$ go to the left child and the others to the right. Each leaf predicts the majority class of the training patients it receives, and the error of a tree is the number of training patients it misclassifies.

(a) (10 points) Grow a tree greedily, as CART does. At the root, choose the split that minimizes the weighted Gini impurity of the two children,

$$\frac{n_{\mathrm{L}}}{n} I_{\mathrm{L}} + \frac{n_{\mathrm{R}}}{n} I_{\mathrm{R}}, \qquad I_t = 1 - \sum_{k \in \{A,B\}} p_{kt}^2,$$

where n_t is the number of patients in child t and p_{kt} the fraction of them in class k. Then split each child of the root the same way, using only the patients in that child. Split both children, so that the tree has three splits and four leaves. Report the impurity of every candidate split at each of the three nodes, the class each leaf predicts, and the error of the tree.

(b) (10 points) Now find the tree with three splits and four leaves that has the smallest error, and give that error. Compare its root split with the ranking of root splits you computed in part (a), under both the Gini impurity and the misclassification error, and explain in a few sentences why growing the tree greedily cannot find it. Finally, can a tree of any depth, still using integer thresholds, have a smaller error than yours?

Hint. Once the root split is fixed, the best split for the left child and the best split for the right child can be chosen separately.

(c) (10 points) Lecture 9 fits optimal trees by solving

$$\min_{T} \quad \text{error}(T) + \alpha \cdot (\text{number of splits in } T),$$

over trees T of depth at most 2, which may use 0, 1, 2 or 3 splits. Here the error is the number of misclassified patients (that is, $\hat{L} = 1$ in the lecture's formulation), and $\alpha \geq 0$ is the complexity parameter.

(i) For every $\alpha \geq 0$, give the optimal tree and its objective value. For which numbers of splits is there no α at which a tree with that many splits is the unique optimum?

(ii) CART instead takes the greedy tree of part (a) and prunes it: for each α, it returns the subtree of that tree, obtained by removing splits, with the smallest value of the same objective. Give CART's tree and objective value for every $\alpha \geq 0$. For which α do CART and the optimal tree reach the same objective value?

(d) The lab panel costs money and time, so the department attaches a cost to each feature: measuring feature j for one patient costs $c_j > 0$. A patient is charged only for the features tested on the path from the root to their leaf, and is charged once for a feature however many times that path tests it, since a lab result, once drawn, can be read again at no extra cost. The department requires that no patient's rule costs more than a budget C.

Recall the notation of the Lecture 9 formulation with parallel splits, for a tree of maximum depth D on p features. $\mathcal{B}$ is the set of branch nodes and $\mathcal{L}$ the set of leaves. Branch node $m \in \mathcal{B}$ carries binaries a_{jm}, $j \in [p]$, and d_m, with

$$\sum_{j=1}^{p} a_{jm} = d_m, \qquad m \in \mathcal{B},$$

so that $d_m = 1$ if node m applies a split and $a_{jm} = 1$ if that split is on feature j. $P_t \subseteq \mathcal{B}$ is the set of ancestors of leaf $t \in \mathcal{L}$.

(i) (6 points) Add linear constraints to the formulation so that, for every leaf, the cost of the features tested on the path to that leaf is at most C. You may introduce auxiliary continuous or binary variables. State what each one does, and say how many variables and constraints you add in terms of p, $|\mathcal{L}|$ and D.

(ii) (4 points) Return to the data of Figure 3.1 with $c_1 = 1$, $c_2 = 3$, $C = 3$ and $\alpha = 0$. Find a tree of depth at most 2 with the smallest error among those that meet the budget, and give that error. A colleague proposes the shorter constraint

$$\sum_{m \in P_t} \sum_{j=1}^{p} c_j \, a_{jm} \leq C, \qquad t \in \mathcal{L},$$

in place of yours. What is the smallest error a tree can reach under the colleague's constraint, and why does it differ from yours?

Question 3: Who is about to cancel? (30 points)

You run an AI-assistant startup that sells subscriptions, billed monthly or annually. To cover the GPU bill, in September 2025 you raised the price from \$20 to \$25 a month and introduced usage caps. Cancellations have gone up since, and your investors are getting concerned.

The retention team can offer each subscriber one free month. An offer to someone who would have stayed costs one month of revenue. However, missing someone who then cancels costs an average of five months of revenue (their average remaining subscription lifetime). For computing costs, assume that an offer always keeps a subscriber who would otherwise have canceled, and that the free month for such a subscriber costs nothing to the company. Company policy forbids targeting offers by age, and the board will only approve a model it can interpret easily.

The file `subscribers.csv`, on Canvas, was extracted from the billing and product databases on 30 June 2026. Each row is a subscriber scored at a monthly snapshot (the 15th of the month) between January 2025 and April 2026. Every active subscriber is scored at each snapshot, so a subscriber has one row a month from the month after sign-up until they cancel.

`subscriber_id`	account number
`snapshot_month`	month of the snapshot
`billing`	monthly or annual
`tenure_months`	months since sign-up
`region`	billing region
`age_band`	age group
`signup_channel`	how the subscriber found the product
`active_days_30d`	days active in the 30 days before the snapshot, as logged by the product
`usage_trend`	messages in the 30 days before the snapshot divided by messages in the 30 days before that
`features_used`	product features used in the 30 days before the snapshot (1 to 6)
`support_tickets_90d`	support tickets opened in the 90 days before the snapshot
`failed_payment_30d`	1 if a payment failed in the 30 days before the snapshot
`days_since_last_login`	days from the last login to the snapshot
`last_invoice_usd`	amount of the most recent invoice, in dollars
`churned_60d`	1 if the subscriber canceled within 60 days of the snapshot

Throughout this question:

- The **test set** is the snapshot months February to April 2026; do not use it for training or tuning. Every earlier month is available for training.
- Use IAI with `random_seed=15095`, and with `criterion=:gini` for the tree learners (CART and OCT).
- The **cost per subscriber** of a model is $(5\,\mathrm{FN} + \mathrm{FP})/n$ on the test set, in months of revenue, where FN counts churners the model does not flag, FP counts stayers it does flag, and n is the number of rows in the test set, one per subscriber per snapshot month.
- There is no starter notebook or Julia template for this question; write the code yourself.

Use of AI. You may use AI assistants to write and debug your code and to help you explore the data. Your interpretation of the results and written answers must be your own.

(a) (10 points) Some columns should not be used in the model. Give a table with a row for each column or group of columns, saying whether you keep it as a feature or drop it, with a one-line reason. Say which snapshot months you train on and why, citing a number or plot from the data.

Then fit an OCT without weights, with `max_depth=3`, `minbucket=50` and `cp=0.001`, on all snapshot months up to January 2026, twice: once on every column except `snapshot_month` and `churned_60d`, and once on the columns you kept. Report each tree's root split and test AUC, and explain the difference.

(b) (10 points) On your training months and columns, give churners weight 5 and everyone else weight 1, and tune four models with `GridSearch`:

- CART (an `OptimalTreeClassifier` with `localsearch=false`), over `max_depth` $\in \{2, 3, 4\}$ and `minbucket` $\in \{25, 50, 100\}$, letting `GridSearch` tune `cp`;
- an OCT, over the same grid;
- a `RandomForestClassifier`, over `max_depth` $\in \{5, 10, 15\}$ and `minbucket` $\in \{1, 25, 100\}$;
- an `XGBoostClassifier`, over `num_round` $\in \{50, 100, 200\}$ and `max_depth` $\in \{2, 4, 6\}$.

In one table, report the selected parameters, test AUC and cost per subscriber for the four models, and the cost of two baselines: making no offers, and making an offer to everyone. Then answer:

- Which term of the OCT objective does the 5:1 cost change?
- With these weights, above what share of churners does a leaf predict churn?
- What does `GridSearch` hold out for validation?
- Is that the right choice for this data?

(c) (10 points) Include the tree you would deploy as a figure, then write bullet points for the board:

1. which model you would deploy, and why;
2. which subscribers it sends offers to;
3. how you checked the age policy.